

Beneficial immunostimulatory effect of short-term Chlorella supplementation: enhancement of Natural Killer cell activity and early inflammatory response (Randomized, double-blinded, placebo-controlled trial)

Background In vitro and animal studies have demonstrated that Chlorella is a potent biological response modifier on immunity. However, there were no direct evidences for the effect of Chlorella supplementation on immune/inflammation response in healthy humans. **Methods** This study was designed for an 8-week randomized, double-blinded, placebo-controlled trial: 5g of Chlorella (n=23) or Placebo (n=28) as form of tablets. Mainly, cytotoxic activities of Natural killer (NK) cells and serum concentrations of interferon- γ , interleukin-1 β and interleukin-12 were measured. **Results** After the 8-week, serum concentrations of interferon- γ ($p<0.05$) and interleukin-1 β ($p<0.001$) significantly increased and that of interleukin-12 ($p<0.1$) tended to increase in the Chlorella group. The increments of these cytokines after the intervention were significantly bigger in the Chlorella group than those in the placebo group. In addition, NK cell activities (%) were significantly increased in Chlorella group, but not in Placebo group. The increments of NK cell activities (%) were also significantly bigger in the Chlorella group than the placebo group. Additionally, changed levels of NK cell activity were positively correlated with those of serum interleukin-1 β ($r=0.280$, $p=0.047$) and interferon- γ ($r=0.271$, $p<0.005$). Significantly positive correlations were also observed among the changed levels of serum cytokines; between interferon- γ and interleukin-1 β ($r=0.448$, $p<0.001$), between interleukin-12 and interleukin-1 β ($r=0.416$, $p=0.003$) and between interleukin-12 and interferon- γ ($r=0.570$, $p<0.001$). **Conclusion** These results may suggest a beneficial immunostimulatory effect of short-term Chlorella supplementation which enhances the NK cell activity and produces interferon- γ and interleukin-12 as well as interleukin-1 β , the Th-1 cell-induced cytokines in healthy people.